

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5093C

5 Credits: 4

Program Elective

Source-grounded

Chemical Engineering

□ To understand the basic units, unit operations & unit process in chemical engineering.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5093C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5093C.pdf from the uploaded official SITTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	5093C
Course title	Chemical Engineering
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4,T:0,P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To understand the basic units, unit operations & unit process in chemical engineering.
- □ To comprehend the mechanism of heat transfer by conduction, convection, radiation
- and evaporation.
- □ To explain and differentiate the mechanism of mass transfer by diffusion, absorption,
- leaching, extraction, adsorption and drying.
- □ To describe the basic principles and types of distillation.

Course outcomes

CO	Official outcome	Study level
CO1	14 Applying process in chemical engineering. Apply mechanism of heat transfer by conduction,	See official syllabus
CO2	convection, radiation to solve problems in chemical 15 Applying engineering. Paraphrase mechanism of mass transfer by diffusion,	See official syllabus
CO3	absorption, leaching, extraction, adsorption and 14 Understanding drying. Generalize the basic principles and types of	See official syllabus
CO4	15 Applying distillation	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	engineering.	4	As specified
Module 2	solve problems in chemical engineering.	4	As specified
Module 3	extraction, adsorption and drying.	4	As specified
Module 4	Generalize the basic principles and types of distillation	4	As specified

MODULE 1

engineering.

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: engineering.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain the various systems of units 3 Understanding Describe unit operations and unit process in the

Explain the various systems of units 3 Understanding Describe unit operations and unit process in the is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the various systems of units 3 Understanding Describe unit operations and unit process in the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the various systems of units 3 understanding describe unit operations and unit process in the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the various systems of units 3 Understanding Describe unit operations and unit process in the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the various systems of units 3 Understanding Describe unit operations and unit process in the definition/meaning . steps, application . Technical terms English- .

▼ 3 Understanding field of chemical engineering Solve problems related to atomic weight,

3 Understanding field of chemical engineering Solve problems related to atomic weight, is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding field of chemical engineering Solve problems related to atomic weight, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding field of chemical engineering solve problems related to atomic weight, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 3 Understanding field of chemical engineering Solve problems related to atomic weight, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding field of chemical engineering Solve problems related to atomic weight, definition/meaning. steps, application. Technical terms English- Malayalam.

▼ molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing

molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, molecular weight and equivalent weight of a solid 4 applying substance. solve problems related to expressing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- $\square\square$ molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing $\square\square\square\square\square\square\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

▼ 4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to

4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of

heat transfer by conduction, convection, radiation to using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying concentration of a solution and gases. units and dimensions - various systems of units - fundamental & derived units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - atomic weight, molecular weight, equivalent weight - normality - molarity - molality - mass percent - mass, volume, mole relation, - relationships for gaseous systems - simple problems. apply mechanism of heat transfer by conduction, convection, radiation to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to
 പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English- പദപരിചയം പദപരിചയം പദപരിചയം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**solve problems in chemical engineering.**

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: solve problems in chemical engineering.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to

Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify different modes of heat transfer 3 understanding apply fourier's law of thermal conduction to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to definition/meaning . steps, application . Technical terms English- .

▼ derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it.

derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it. is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, derive heat transfer equations for various solid 8 applying surfaces and solve problems related to it. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നു തിരിച്ചറിയുന്ന ചൂട് കൈമാറ്റ സമവാക്യങ്ങൾ വ്യത്യസ്ത ഘനങ്ങളിലെ ഉപരിതലങ്ങളിൽ പ്രയോഗിച്ച് പ്രശ്നങ്ങൾ പരിഹരിക്കുക.

പ്രശ്നങ്ങൾ പരിഹരിക്കാൻ നിങ്ങൾക്ക് definition/meaning നൽകേണ്ടതാണ്. നിങ്ങൾക്ക് പ്രശ്നങ്ങൾ പരിഹരിക്കാൻ, steps, application നൽകേണ്ടതാണ്. Technical terms English-ൽ നിങ്ങൾക്ക് നൽകേണ്ടതാണ്.

▼ Illustrate various heat exchangers used in industry

Illustrate various heat exchangers used in industry is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate various heat exchangers used in industry using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate various heat exchangers used in industry should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate various heat exchangers used in industry means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate various heat exchangers used in industry □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

▼ **Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching,**

Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical

evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the types of evaporators used in industry. 2 understanding conduction - fourier's law - derivation - problems - conduction through plane wall, cylindrical wall - derivations - problems -, convection - concept of natural convection, radiation idea of black body - gray body - emissivity - absorptivity, heat exchanger patterns - shell and tube - double pipe heat exchangers - evaporators - horizontal and vertical evaporators. paraphrase mechanism of mass transfer by diffusion, absorption, leaching, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, definition/meaning steps, application Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

extraction, adsorption and drying.

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: extraction, adsorption and drying.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding equipment's.

4 Understanding equipment's. is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding equipment's. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding equipment's. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding equipment's. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding equipment's.

പ്രദേശം definition/meaning പ്രദേശം. പ്രദേശം പ്രദേശം പ്രദേശം, steps, application പ്രദേശം പ്രദേശം. Technical terms English- പ്രദേശം പ്രദേശം.

▼ Generalize leaching technique

Generalize leaching technique is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Generalize leaching technique using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, generalize leaching technique should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Generalize leaching technique means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എം Generalize leaching technique
 എങ്ങനെയാണ് definition/meaning എങ്ങനെ എങ്ങനെ
 എങ്ങനെ, steps, application എങ്ങനെ എങ്ങനെ എങ്ങനെ. Technical terms
 English-ൽ എങ്ങനെ എങ്ങനെ എങ്ങനെ.

▼ Describe extraction and extraction equipment's

Describe extraction and extraction equipment's is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe extraction and extraction equipment's using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe extraction and extraction equipment's should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe extraction and extraction equipment's means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നു വിവരിക്കുന്ന എക്സ്ട്രാക്ഷൻ ഉപകരണങ്ങളുടെ നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ എഴുതുക.

▼ **Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier**

Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize drying equipment's used in industry 4 understanding diffusion - fick's law - law of conservation of mass - henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of bollman extractor - extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - drying - applications - working of tray and spray drier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier
 definition/meaning .
 steps, application
 Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Generalize the basic principles and types of distillation

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Generalize the basic principles and types of distillation
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive

Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, differentiate various types of distillation. 3 understanding apply raoul's law and daltons law to derive should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive definition/meaning.
definition/meaning.
steps, application. Technical terms English-
.

▼ equation connecting mole fraction and vapor of a 5 Applying binary mixtures

equation connecting mole fraction and vapor of a 5 Applying binary mixtures is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify equation connecting mole fraction and vapor of a 5 Applying binary mixtures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, equation connecting mole fraction and vapor of a 5 applying binary mixtures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What equation connecting mole fraction and vapor of a 5 Applying binary mixtures means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

▼ Derive equation for relative volatility

Derive equation for relative volatility is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Derive equation for relative volatility using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, derive equation for relative volatility should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Derive equation for relative volatility means in the official course context

Malayalam support: Module 4- σ Derive equation for relative volatility σ_{12} definition/meaning σ_{12} . σ_{12} σ_{12} σ_{12} , steps, application σ_{12} σ_{12} σ_{12} . Technical terms English- σ σ_{12} σ_{12} .

▼ **Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar>**

Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar> is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgary> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare azeotropic and extractive distillation 3 understanding distillation - types - simple and steam distillation -fractional distillation - raoult's law - daltons law - relative volatility - derivation - simple problems - minimum and maximum boiling azeotropes -comparison - azeotropic and extractive distillation. text / reference: t/r book title/author process calculations for chemical engineers - chem. engg., edn development t1 centre, iit madras r1 process calculations - v. venkataramani, n. anantharaman introduction

to chemical engineering - walter.l.badger (publisher: mcgraw-hill r2 inc. us; international 1955 edition) r3 unit operations - mccabe smith (publisher - mcgraw-hill education; 7edition) mass transfer - robert e treybal (publisher - mcgraw-hill book company; 3rd r4 edition) online resources: sl.no website link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 https://www.researchgate.net 2 http://cc.sit.edu.cn 3 http://www.chem.ucalgar means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgary.ca/chem1/definition/meaning> definition/meaning. steps, application. Technical terms English- Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

- ▶ **Define or explain Explain the various systems of units 3 Understanding Describe unit operations and unit process in the. (3 marks)**
- ▶ **Define or explain 3 Understanding field of chemical engineering Solve problems related to atomic weight,. (3 marks)**
- ▶ **Define or explain molecular weight and equivalent weight of a solid 4 Applying substance. Solve problems related to expressing. (3 marks)**
- ▶ **Define or explain 4 Applying concentration of a solution and gases. Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems. Apply mechanism of heat transfer by conduction, convection, radiation to. (3 marks)**

Module 2

- ▶ **Define or explain Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to. (3 marks)**

► **Define or explain derive heat transfer equations for various solid 8 Applying surfaces and solve problems related to it.. (3 marks)**

► **Define or explain Illustrate various heat exchangers used in industry. (3 marks)**

► **Define or explain Discuss the types of evaporators used in industry. 2 Understanding Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers - Evaporators - horizontal and vertical evaporators. Paraphrase mechanism of mass transfer by diffusion, absorption, leaching,. (3 marks)**

Module 3

► **Define or explain 4 Understanding equipment's.. (3 marks)**

► **Define or explain Generalize leaching technique. (3 marks)**

► **Define or explain Describe extraction and extraction equipment's. (3 marks)**

► **Define or explain Summarize drying equipment's used in industry 4 Understanding Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier. (3 marks)**

Module 4

► **Define or explain Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive. (3 marks)**

► **Define or explain equation connecting mole fraction and vapor of a 5 Applying binary mixtures. (3 marks)**

► **Define or explain Derive equation for relative volatility. (3 marks)**

► **Define or explain Compare Azeotropic and Extractive Distillation 3 Understanding Distillation - types - Simple and Steam distillation - fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation. Text / Reference: T/R Book Title/Author Process Calculations for chemical engineers - Chem. Engg., Edn Development T1 Centre, IIT Madras R1 Process Calculations - V. Venkataramani, N. Anantharaman Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill R2 Inc. US; International 1955 Edition) R3 Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition) Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd R4 edition) Online Resources: Sl.No Website Link 1 <https://www.researchgate.net> 2 <http://cc.sit.edu.cn> 3 <http://www.chem.ucalgar>. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the various systems of units 3 Understanding Describe unit operations and unit process in the.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Identify different modes of heat transfer 3 Understanding Apply Fourier's law of thermal conduction to.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Understanding equipment's..**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Differentiate various types of distillation. 3 Understanding Apply Raoul's law and Daltons law to derive.**

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.researchgate.net> <http://cc.sit.edu.cn>
<http://www.chem.ucalgary>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5093C. Verify institutional updates against the current official curriculum.